

Binjie Zhang

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Research focus: LLM agents, multimodal understanding, and reliable model adaptation.

EDUCATION

National University of Singapore (NUS) <i>Ph.D. in Computer Science – Advisor: Prof. Mike Zheng Shou</i>	Singapore Jan 2023 – Jan 2027 (expected)
Tsinghua University (THU) <i>M.Eng. in Computer Science and Technology – Advisor: Prof. Chun Yuan</i>	Beijing, China Aug 2019 – Jul 2022
East China University of Science and Technology (ECUST) <i>B.Eng. in Information Engineering – GPA 3.77/4.00, Rank 1/92</i>	Shanghai, China Aug 2015 – Jul 2019

SELECTED PUBLICATIONS

4 first/co-first-author conference papers; 1 additional preprint.

“ReGRPO: Reflection-Augmented Policy Optimization for Tool-Using Agents.” *ECCV*, 2026.
“Darwinian Model Upgrades: Model Evolving with Selective Compatibility.” *AAAI*, 2023 (**co-first author**).

“Towards Universal Backward-Compatible Representation Learning.” *IJCAI*, 2022 (**Long Oral**).
“Hot-Refresh Model Upgrades with Regression-Alleviating Compatible Training in Image Retrieval.” *ICLR*, 2022.
“TaCA: Upgrading Your Visual Foundation Model with Task-agnostic Compatible Adapter.” *arXiv preprint*, 2023.

Under review: SGRC, StateTrackBench, PathComp (*ICLR* 2027); Keep or Reset / POU-OPD (*AAAI*).

RESEARCH & WORK EXPERIENCE

TikTok Search (ByteDance) Singapore
AI Research Intern – Agentic Search and Agent Model Training 2026 – Present
○ **Agentic Search:** led Auto-Skill evolution and scenario-specific prompting; converted failure feedback into reusable agent policies for planning, tool use, evidence integration, and answer delivery.
○ **Model learning** (ongoing): developing observation-faithful training data, tool-use learning objectives, and evaluation that distinguishes agent decisions from delivered answer quality.

TikTok Shop (ByteDance) Singapore
AI Research Intern – Led three agent systems into production within five months Jan – Jun 2026
○ **FIRE – Agent Harness:** designed a reusable, skill-based execution framework with semantic routing, progressive context loading, memory access, and controlled tool execution.
○ **EvoA – Model Iteration:** built a diagnosis–experiment–reflection loop with reusable experiment memory and staged validation; reduced human-audit rate by **about 4.6% relatively** in an online comparison.
○ **APA – Risk Discovery:** designed evidence-grounded discovery with related-entity expansion; increased weekly **candidate throughput by about 17×** versus earlier workflow versions.

Tencent ARC Lab / PCG Shenzhen, China
AI Research Intern – Compatible Learning and Cross-Modal Video Understanding 2019 – 2023
○ **Research to deployment:** led first/co-first-author work on compatible retrieval and hot-refresh upgrades; reduced feature-refresh costs and accelerated model deployment while preserving retrieval behavior.
○ Built contrastive learning and video–text retrieval pipelines; contributed to work recognized by the Tencent Technology Breakthrough Award and SZCCF Science and Technology Award (2022).

HONORS AND AWARDS

- Tencent Technology Breakthrough Award; SZCCF Science and Technology Award 2022
- Excellent Master’s Graduate in Beijing; Outstanding Master’s Graduation Thesis 2022
- National Scholarship for Undergraduates (twice), Ministry of Education of China 2017, 2016

TECHNICAL SKILLS

Methods: agent harness, memory, context management, tool use, reinforcement learning, multimodal and compatible representation learning.

Tools: Python, PyTorch, Hugging Face, DeepSpeed, vLLM, MCP; distributed training, data pipelines, and sandboxed execution.